

Curtin University, Semester 2, 2026**ECOM5005 Tutorial 8 Solution***Hypothesis Testing: One-sample Tests***Learning Outcomes**

1. Identify the basic principles of hypothesis testing.
2. Explain the assumptions of each one-sample testing procedure and the consequences of serious violations.
3. Use critical values and p-values to test a population mean or proportion.
4. Interpret Type I and Type II errors and distinguish statistical significance from practical significance.
5. Apply one-sample hypothesis tests in Excel and communicate a managerial conclusion.

PART 1 – Detailed solutions**Question 1**

For the hypothesis test below, where the decision is whether to reject H_0 , then a ____ error could have occurred.

$H_0: \mu \geq 90$, $H_1: \mu < 90$, $\alpha = 10\%$

Critical value = -1.28; Test value = -2.3; Decision: Reject H_0 ; Conclusion: The mean is less than 90.

A. Type I B. Type II C. None

Detailed solution

- The decision shown is to reject H_0 .
- Whenever H_0 is rejected, the possible mistake is a Type I error: rejecting H_0 even though H_0 is actually true.
- The probability of making a Type I error is the significance level α . Here $\alpha = 0.10$, so the testing procedure accepts a 10% Type I error rate in repeated use when the null is true.
- A Type II error is only possible when we fail to reject H_0 while H_0 is false. That is not the decision made here.

Answer

A. Type I error.

In context: we could conclude that the population mean is below 90 when, in truth, $\mu \geq 90$.

Question 2

A survey claims that 9 out of 10 doctors recommend aspirin for their patients with headaches. To test this claim against the alternative that the actual proportion of doctors who recommend aspirin is less than 0.90, a random sample of 100 doctors results in 83 who indicate that they recommend aspirin. The value of the test statistic in this problem is approximately equal to

- A. -0.07 B. -2.33 C. -1.86 D. -4.12

Detailed solution

6. Identify the population parameter. Let π be the true proportion of doctors who recommend aspirin.
7. State the hypotheses: $H_0: \pi = 0.90$ (equivalently $\pi \geq 0.90$ for the lower-tail test) and $H_1: \pi < 0.90$.
8. Compute the sample proportion: $\hat{p} = 83/100 = 0.83$.
9. Check the normal approximation under H_0 : $n\pi_0 = 100(0.90) = 90$ and $n(1-\pi_0) = 100(0.10) = 10$. Both are at least 5.
10. Calculate the null standard error.

$$SE_0 = \sqrt{[\pi_0(1-\pi_0)/n]} = \sqrt{[(0.90)(0.10)/100]} = 0.0300$$

11. Calculate the z statistic.

$$z = (\hat{p} - \pi_0) / SE_0 = (0.83 - 0.90) / 0.0300 = -2.3333$$

Answer

The test statistic is approximately -2.33, so the correct choice is B.

For reference, the lower-tail p-value is approximately 0.0098.

Question 3

A major pet store chain is considering opening a new store in an area that currently does not have any such stores. The chain will open if there is evidence that more than 5,000 of the 20,000 households in the area own pets. It conducts a telephone poll of 300 randomly selected households in the area and finds that 96 have pets.

(1) State the test of hypothesis that is of interest to the pet store chain.

- A. $H_0: \pi \leq 5,000$ vs. $H_1: \pi > 5,000$ B. $H_0: \pi \leq 0.25$ vs. $H_1: \pi > 0.25$ C. $H_0: \pi \leq 0.32$ vs. $H_1: \pi > 0.32$ D. $H_0: \pi \leq 5,000$ vs. $H_1: \pi > 5,000$

(2) The value of the test statistic in this problem is approximately equal to

- A. 1.94 B. 1.30 C. 2.60 D. 2.80

(3) The p-value associated with the test statistic in this problem is approximately equal to

- A. 0.0100 B. 0.0026 C. 0.0051 D. 0.0013

(4) The decision on the hypothesis test using a 5% level of significance is

- A. To accept H_0 in favour of H_1 B. To fail to reject H_0 in favour of H_1 C. To reject H_0 in favour of H_1 D. We cannot tell what the decision should be from the information given.

(5) The chain's conclusion from the hypothesis test using a 5% level of significance is

- A. To delay opening a new store until additional evidence is collected B. Not to open a new store C. To open a new store D. We cannot tell what the decision should be from the information given.

Detailed solution

12. Translate the business threshold into a population proportion. The store needs evidence that more than 5,000 of 20,000 households own pets, so $\pi_0 = 5000/20000 = 0.25$.
13. State the hypotheses: $H_0: \pi \leq 0.25$ and $H_1: \pi > 0.25$. This is an upper-tail test. Therefore part (1) is B.
14. Compute the sample proportion: $\hat{p} = 96/300 = 0.32$.
15. Check the normal approximation: $n\pi_0 = 300(0.25) = 75$ and $n(1-\pi_0) = 300(0.75) = 225$, both comfortably above 5.
16. Compute the null standard error and test statistic.

$$SE_0 = \sqrt{[(0.25)(0.75)/300]} = 0.0250$$

$$z = (0.32 - 0.25)/0.0250 = 2.80$$

- Part (2): $z \approx 2.80$, so the correct choice is D.
17. For an upper-tail test, the p-value is $P(Z \geq 2.80) \approx 0.0026$.
 - Part (3): p-value ≈ 0.0026 , so the correct choice is B.
 18. Compare p-value with $\alpha = 0.05$. Because $0.0026 < 0.05$, reject H_0 .
 - Part (4): reject H_0 in favour of H_1 , so the correct choice is C.
 19. Managerial conclusion: there is statistically significant evidence that more than 25% of households own pets. This supports the chain's stated condition for opening the store.
 - Part (5): based on the stated decision rule, open the new store, so the correct choice is C.

Answer summary

- (1) B: $H_0: \pi \leq 0.25$ vs. $H_1: \pi > 0.25$
- (2) D: $z = 2.80$
- (3) B: $p \approx 0.0026$
- (4) C: reject H_0
- (5) C: open the new store

Real-life problems: detailed solutions

Question 4 – EV battery range and a Z test

An electric-vehicle manufacturer advertises that the mean highway range of a new battery pack is 400 km. Long-run engineering records indicate that the population standard deviation is $\sigma = 6$ km. An independent random sample of 64 production packs has a mean range of 402.1 km.

- (a) State H_0 and H_1 for testing whether the true mean range differs from 400 km.
- (b) At $\alpha = 0.01$, use the critical-value approach to perform the test.
- (c) Calculate the two-tail p-value and make the same decision using the p-value approach.
- (d) Construct the corresponding 99% confidence interval for μ and explain its connection to the test.
- (e) Explain a Type I error in this context and comment on statistical significance versus practical significance.

Detailed solution

20. Parameter and hypotheses. Let μ be the true mean highway range of the new battery pack. Because the question asks whether the mean differs from 400 km, use a two-tail test: $H_0: \mu = 400$; $H_1: \mu \neq 400$.

21. Choose the test. The population standard deviation is given as $\sigma = 6$ km, so use a one-sample Z test.
 22. Compute the standard error and Z statistic.

$$SE = \sigma/\sqrt{n} = 6/\sqrt{64} = 0.75 \text{ km}$$

$$z = (402.1 - 400)/0.75 = 2.80$$

23. Critical-value approach. For $\alpha = 0.01$ in a two-tail test, $\alpha/2 = 0.005$ in each tail and the critical values are approximately ± 2.576 .

- Decision: $|2.80| > 2.576$, so reject H_0 .

24. p-value approach. The two-tail p-value is $2P(Z \geq 2.80) \approx 0.0051$.

- Because $0.0051 < 0.01$, reject H_0 again.

25. 99% confidence interval.

$$402.1 \pm 2.576(0.75) = 402.1 \pm 1.932$$

$$400.168 \text{ km} \leq \mu \leq 404.032 \text{ km}$$

- The interval does not contain the hypothesised value 400 km, so the 99% confidence-interval conclusion agrees with the two-tail test at $\alpha = 0.01$.

26. Type I error and practical significance.

- A Type I error would be concluding that the true mean range differs from 400 km when the true mean actually is 400 km.
- The estimated difference is only 2.1 km. It is statistically detectable because the standard error is small, but management should separately decide whether a 2.1 km change is large enough to matter operationally or commercially.

Final answer

$z = 2.80$; critical values = ± 2.576 ; $p \approx 0.0051$. Reject H_0 at the 1% level.

99% CI: approximately (400.17 km, 404.03 km).

Question 5 – AI helpdesk improvement and a one-sample t test

A SaaS company has redesigned its AI helpdesk workflow. Historically, the mean time from a customer opening a ticket to a human handoff was 12.0 minutes. After the redesign, a random sample of 36 tickets has a mean handoff time of 11.1 minutes and a sample standard deviation of 2.4 minutes. Management wants evidence that the redesign has reduced the mean handoff time.

(a) State H_0 and H_1 and identify whether this is a lower-tail, upper-tail or two-tail test.

(b) At $\alpha = 0.05$, calculate the t statistic, degrees of freedom and critical value.

(c) Calculate the p-value and make a statistical decision.

(d) State the managerial conclusion, explain a Type I error, and comment on whether the estimated 0.9-minute reduction is practically meaningful.

Detailed solution

27. Let μ be the post-redesign population mean handoff time. Management wants evidence of a reduction: $H_0: \mu \geq 12.0$ versus $H_1: \mu < 12.0$. This is a lower-tail test.

28. Because σ is unknown and the sample standard deviation $S = 2.4$ is used, apply a one-sample t test. Degrees of freedom: $df = 36 - 1 = 35$.

$$SE = S/\sqrt{n} = 2.4/\sqrt{36} = 0.40 \text{ minutes}$$

$$t = (11.1 - 12.0)/0.40 = -2.25$$

29. For $\alpha = 0.05$ and $df = 35$, the lower-tail critical value is approximately -1.690 .
- Since $-2.25 < -1.690$, the test statistic is in the rejection region. Reject H_0 .
30. The lower-tail p -value is $P(T_{35} \leq -2.25) \approx 0.0154$.
- Because $0.0154 < 0.05$, the p -value approach also rejects H_0 .
31. Managerial conclusion: there is sufficient evidence at the 5% level that the redesign reduced the mean handoff time below 12 minutes.
32. A Type I error would be concluding that the redesign reduced the mean handoff time when the true mean is still 12 minutes or higher.
33. Practical significance: the estimated reduction is 0.9 minute, or 7.5% of the 12-minute benchmark. Whether that is economically important depends on ticket volume, staffing cost and service-level objectives.

Final answer

$t = -2.25$, $df = 35$, critical $t \approx -1.690$, $p \approx 0.0154$.

Reject H_0 at $\alpha = 0.05$; there is evidence that mean handoff time has decreased.

Question 6 – Streaming conversion and a proportion test

A streaming platform historically converts 30% of free trials to paid subscriptions. After redesigning the onboarding process, 139 of 400 randomly selected new trials convert to paid subscriptions. The product team wants to know whether the conversion rate has increased.

- State H_0 and H_1 and check the normal-approximation conditions for a one-sample proportion test.
- At $\alpha = 0.05$, calculate the z statistic and the upper-tail critical value.
- Calculate the p -value and state the decision at $\alpha = 0.05$.
- Would the decision change at $\alpha = 0.01$? Explain.
- Explain a Type I error in this context and distinguish statistical significance from business significance.

Detailed solution

34. Let π be the post-redesign population conversion rate. The business question is whether conversion increased: $H_0: \pi \leq 0.30$ versus $H_1: \pi > 0.30$. This is an upper-tail test.
35. Check the normal approximation using the null value $\pi_0 = 0.30$: $n\pi_0 = 400(0.30) = 120$ and $n(1-\pi_0) = 400(0.70) = 280$. Both exceed 5.
36. Compute $\hat{p} = 139/400 = 0.3475$ and the null standard error.

$$SE_0 = \sqrt{[(0.30)(0.70)/400]} = 0.02291$$

$$z = (0.3475 - 0.30)/0.02291 = 2.073$$

37. At $\alpha = 0.05$, the upper-tail critical z value is 1.645. Because $2.073 > 1.645$, reject H_0 .
38. The upper-tail p -value is $P(Z \geq 2.073) \approx 0.0191$. Because $0.0191 < 0.05$, reject H_0 .
39. At $\alpha = 0.01$, the p -value 0.0191 is greater than 0.01, so do not reject H_0 . The evidence is strong enough for 5%, but not for the more stringent 1% threshold.

40. A Type I error would be concluding that the redesigned onboarding raises conversion above 30% when the true conversion rate is actually 30% or lower.
41. Business significance: the sample conversion rate is 34.75%, an increase of 4.75 percentage points. The team should consider the revenue value of that increase and whether the random sample represents future users.

Final answer

$z \approx 2.073$; $p \approx 0.0191$.

Reject H_0 at $\alpha = 0.05$, but do not reject H_0 at $\alpha = 0.01$.

PART 2 – Data exercise: detailed Excel solution

Question 7 – Consumer credit-card expenditure

This exercise uses Bill Greene's credit-card expenditure data supplied in the Excel workbook ECOM5005_Tutorial_8_Consumer_Credit_Card_Data.xlsx. The workbook contains 72 observations. AVGEXP is average monthly credit-card expenditure; OWNRENT equals 1 for a homeowner and 0 for a renter.

- (a) Open the workbook and inspect the Read_Me and Credit_Card_Data sheets. Do not sort or delete the raw data.
- (b) For AVGEXP, calculate n, mean, median, sample standard deviation, minimum and maximum. Create a histogram and briefly describe the shape of the distribution.
- (c) A planning benchmark states that mean monthly credit-card expenditure is \$200. At $\alpha = 0.05$, test $H_0: \mu = 200$ against $H_1: \mu \neq 200$ using a one-sample t test. Show the hypotheses, test statistic, degrees of freedom, critical value, p-value, decision and business conclusion.
- (d) Construct a 95% confidence interval for μ and explain why it should lead to the same decision as the two-tail test in part (c).
- (e) Create a frequency table and column chart for HOMEOWNER. A planning assumption is that 50% of cardholders own their home. At $\alpha = 0.05$, test $H_0: \pi \geq 0.50$ against $H_1: \pi < 0.50$. Check the normal-approximation conditions and report z, critical z, p-value, decision and conclusion.
- (f) Comment on the assumptions and limitations of the analysis, including the strong skewness in expenditure, sample representativeness, and the difference between statistical and practical significance.

Dataset source: Statsmodels "Bill Greene's credit scoring data", sourced from William Greene's Econometric Analysis.
<https://www.statsmodels.org/dev/datasets/generated/ccard.html>

A. Before you start in Excel

42. Download and open "Consumer_Credit_Card_Data.xlsx".
43. Open the Read_Me sheet first. It explains the variables and the source.
44. Open Credit_Card_Data. Row 1 contains variable names and rows 2–73 contain 72 observations. Do not delete or sort the raw data.
45. Open Analysis_Workspace. Enter formulas in the "Your result" column (column D). This keeps the raw data separate from your calculations.

B. Descriptive statistics for AVGEXP

Statistic	Excel formula	Final result
Sample size n	=COUNT(Credit_Card_Data!B2:B73)	72
Mean	=AVERAGE(Credit_Card_Data!B2:B73)	\$262.53
Median	=MEDIAN(Credit_Card_Data!B2:B73)	\$158.32
Sample standard deviation	=STDEV.S(Credit_Card_Data!B2:B73)	\$318.05
Minimum	=MIN(Credit_Card_Data!B2:B73)	\$9.58
Maximum	=MAX(Credit_Card_Data!B2:B73)	\$1,898.03
Homeowners	=COUNTIF(Credit_Card_Data!G2:G73,"Owns home")	27
Homeowner proportion	=D10/D4	0.3750 = 37.50%

- The mean (\$262.53) is much larger than the median (\$158.32), and the maximum is very large. This already suggests a strongly right-skewed expenditure distribution.

C. Create the histogram in Excel

46. Go to Credit_Card_Data and select cells B1:B73, including the AVGEXP heading.
47. Click Insert → Insert Statistic Chart → Histogram.
48. Click the horizontal axis, right-click and choose Format Axis.
49. Set a convenient Bin width such as 200. This groups expenditures into intervals such as \$0–\$200, \$200–\$400, and so on.
50. Add the chart title “Distribution of Average Monthly Credit-Card Expenditure”. Add axis titles if your Excel version allows them.
51. Your histogram should show many observations at relatively low expenditure and a long right tail, including a few very large expenditures.

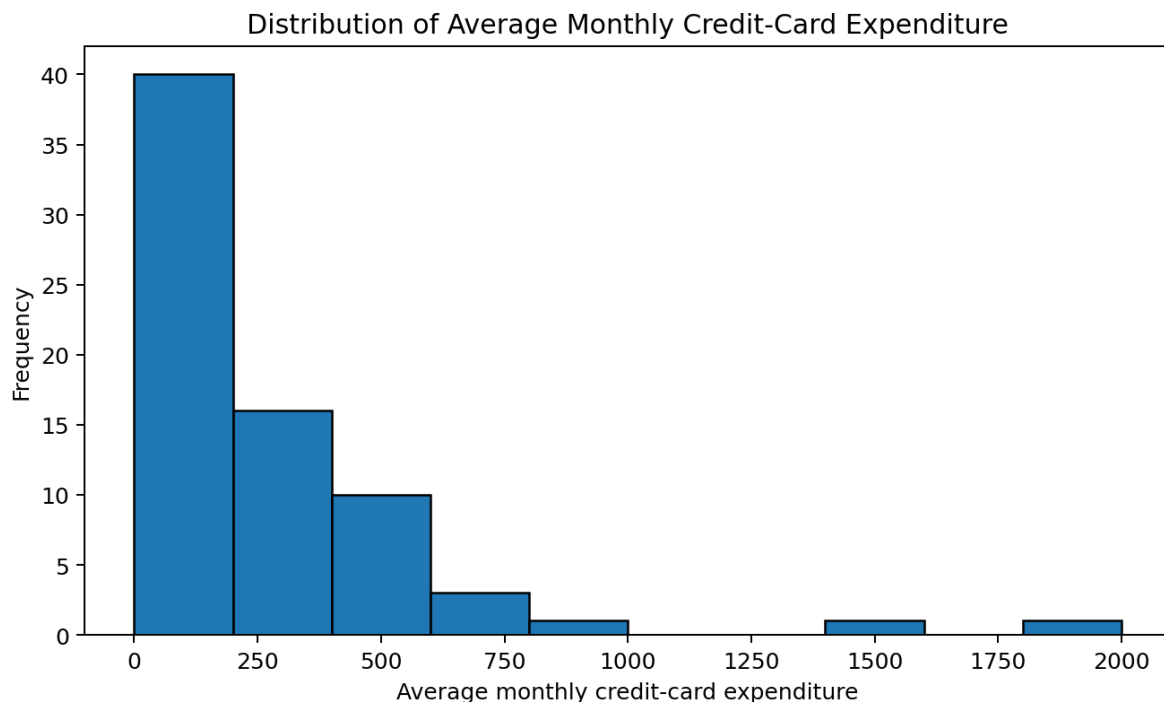


Figure 1: Histogram of average monthly credit-card expenditure (72 observations)

D. One-sample t test for mean monthly expenditure

- Business benchmark: \$200 per month.

$$H_0: \mu = 200 \quad H_1: \mu \neq 200 \quad \alpha = 0.05$$

52. Because the population standard deviation σ is not known, use a one-sample t test. The sample size is $n = 72$, so $df = 71$.
53. In Analysis_Workspace, enter 200 in D12 and 0.05 in D13. Then enter the following formulas:

Cell	What it calculates	Formula	Result
D14	Standard error	=D7/SQRT(D4)	37.4822
D15	t statistic	=(D5-D12)/D14	1.6683
D16	Degrees of freedom	=D4-1	71
D17	Two-tail critical t	=T.INV.2T(D13,D16)	1.9939
D18	Two-tail p-value	=T.DIST.2T(ABS(D15),D16)	0.0997
E19	95% CI lower	=D5-D17*D14	\$187.79
F19	95% CI upper	=D5+D17*D14	\$337.27

54. Critical-value decision: reject H_0 only if $t < -1.9939$ or $t > 1.9939$. Here $t = 1.6683$, which lies in the non-rejection region. Do not reject H_0 .
55. p-value decision: $p = 0.0997 > 0.05$. Again, do not reject H_0 .
56. Managerial conclusion: the sample mean is above \$200, but the data do not provide sufficient evidence at the 5% level to conclude that the population mean monthly expenditure differs from \$200.

Mean-test result

$t(71) = 1.6683$, $p = 0.0997$. Do not reject H_0 at $\alpha = 0.05$.

95% CI for μ : approximately \$187.79 to \$337.27. Because \$200 lies inside the interval, the confidence-interval conclusion agrees with the two-tail hypothesis test.

E. Homeownership frequency table and chart

- Homeowners: 27
 - Renters: 45
 - Sample homeowner proportion: $27/72 = 0.375 = 37.5\%$.
57. In an empty area of Analysis_Workspace, type Homeowner in one cell and Renter below it.
58. Next to Homeowner enter =COUNTIF(Credit_Card_Data!G2:G73,"Owns home").
59. Next to Renter enter =COUNTIF(Credit_Card_Data!G2:G73,"Rents").
60. Select the two labels and counts, then choose Insert → Column or Bar Chart → Clustered Column.

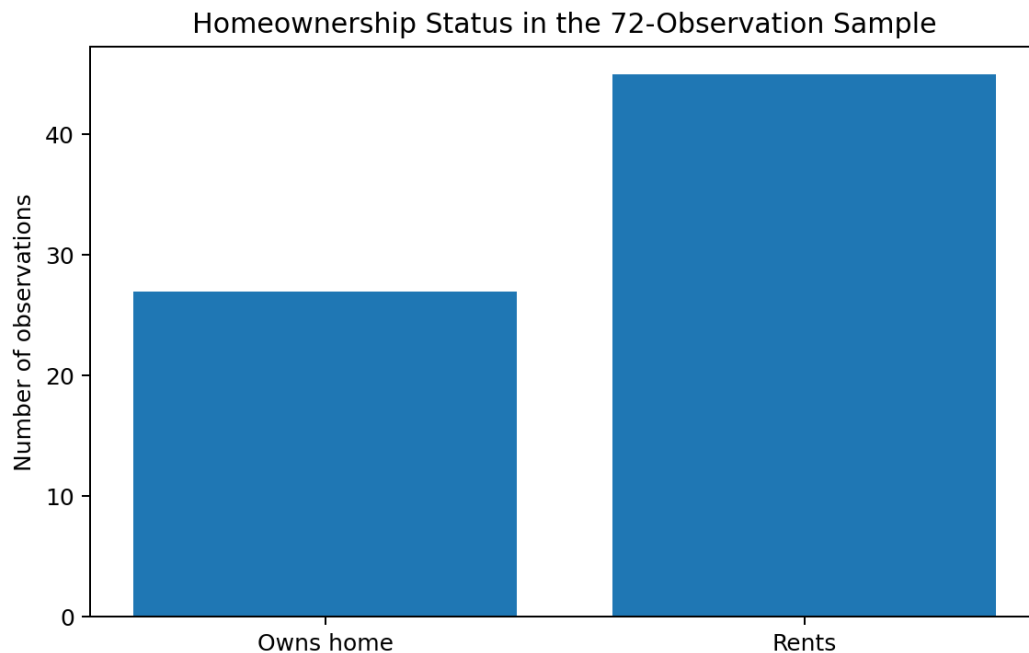


Figure 2: Homeownership status in the sample

F. One-sample proportion test for homeownership

$$H_0: \pi \geq 0.50 \quad H_1: \pi < 0.50 \quad \alpha = 0.05$$

61. Check the normal approximation under H_0 : $n\pi_0 = 72(0.50) = 36$ and $n(1-\pi_0) = 72(0.50) = 36$. Both are well above 5.
62. Enter $\pi_0 = 0.50$ in D20 and $\alpha = 0.05$ in D21. Then use:

Cell	What it calculates	Formula	Result
D22	Null standard error	<code>=SQRT(D20*(1-D20)/D4)</code>	0.05893
D23	z statistic	<code>=(D11-D20)/D22</code>	-2.1213
D24	Lower-tail critical z	<code>=NORM.S.INV(D21)</code>	-1.6449
D25	Lower-tail p-value	<code>=NORM.S.DIST(D23,TRUE)</code>	0.01695

63. Critical-value decision: $-2.1213 < -1.6449$, so reject H_0 .
64. p-value decision: $0.01695 < 0.05$, so reject H_0 .
65. Managerial conclusion: the data provide evidence that the population proportion of homeowners represented by this sample is below 50%.

Proportion-test result

$z = -2.1213$, $p \approx 0.01695$. Reject H_0 at $\alpha = 0.05$.

The sample homeowner rate is 37.5%, which is statistically below the 50% planning benchmark.

G. Assumptions, limitations and interpretation

- The observations should be treated as an appropriate sample from the target population. Statistical inference cannot repair a biased or non-representative sampling process.
- AVGEXP is strongly right-skewed and contains very large observations. With $n = 72$, the sampling distribution of the mean is more stable than the raw data, but the sample mean can still be sensitive to extreme values. The histogram should therefore be reported, not hidden.
- The one-sample proportion test requires a binary outcome and sufficiently large expected counts under H_0 ; here both expected counts are 36.
- A statistically significant result does not automatically imply a large or economically important effect. Conversely, a non-significant result does not prove that the null hypothesis is true.
- Choose α and the direction of the test before inspecting the result. Do not switch from a two-tail test to a one-tail test simply to obtain significance.
- All relevant results—including p-values, effect sizes, uncertainty and unusual observations—should be reported transparently.

Part 2 final results at a glance

Analysis	Key statistic	p-value	Decision at 5%	Business interpretation
Mean AVGEXP vs. \$200	$t = 1.6683$	0.0997	Do not reject H_0	Insufficient evidence that mean monthly expenditure differs from \$200.
Homeowner rate vs. 50%	$z = -2.1213$	0.01695	Reject H_0	Evidence that the homeowner proportion is below 50%.

PART 3 – Pearson MyLab

1. Please ensure that you have completed Quiz 4.

Note: If you are having issues with obtaining access to MyLab, please re-read the instructions (available in the welcome topic). If you are still unsuccessful, please send an email to the teaching team explaining the issue. It is important that we complete MyLab tasks on a weekly basis since the final e-test in this unit might be completed using MyLab.